DP Biology – B1.1

Carbohydrates and lipids

- How do differences in form give carbohydrates and lipids different functions?
- In what ways do lipids and carbohydrates differ as energy stores?

SL statements B1.1.1–13. Textbook pp. 181–204.

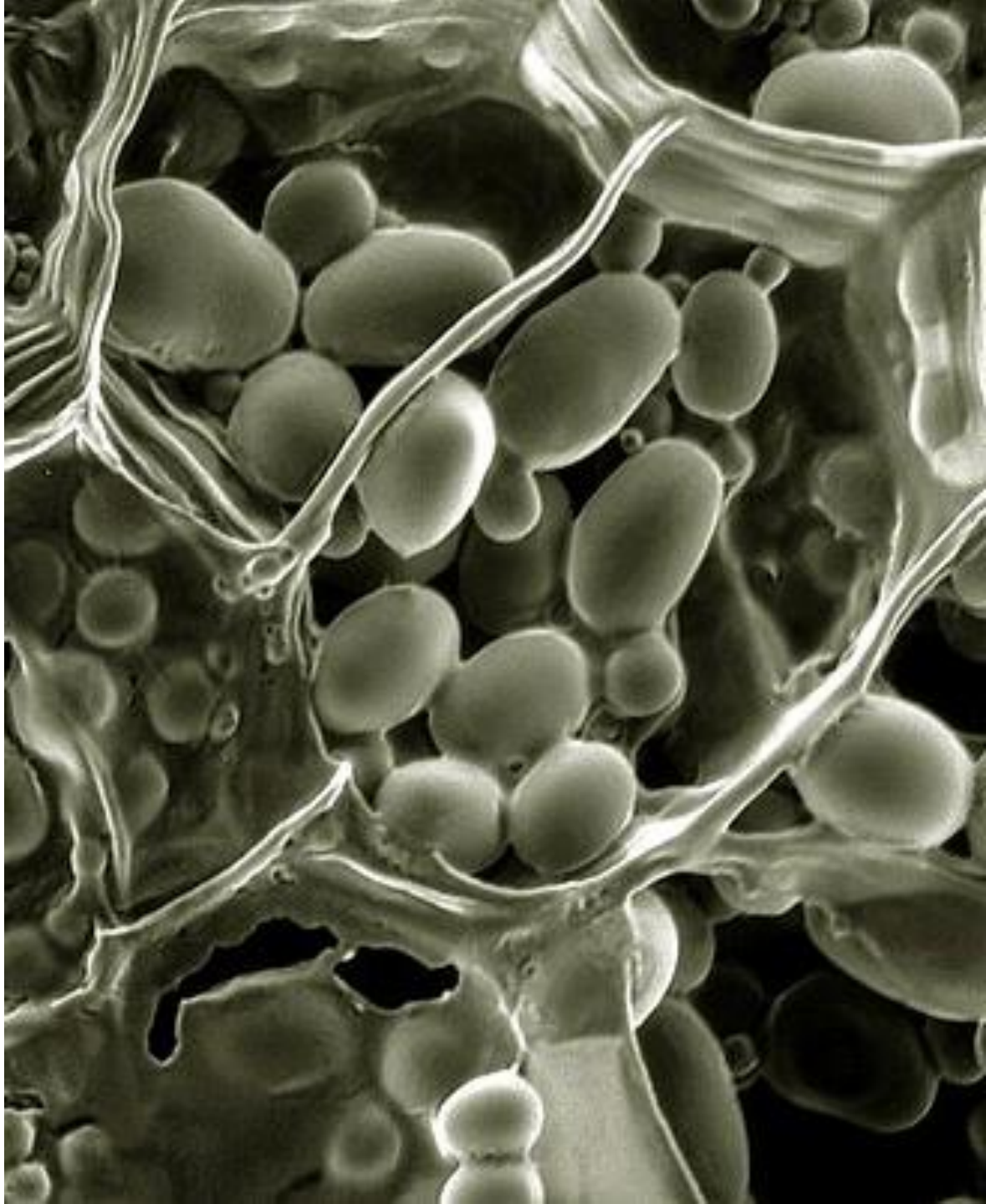
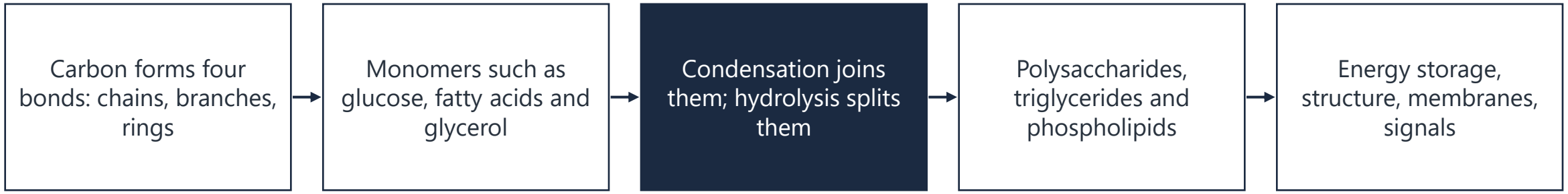


Image: starch grains from a potato, electron micrograph. Philippa Uwins, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can…
1	B1.1.1–3	Explain why carbon forms so many compounds, and how condensation and hydrolysis build and break polymers
2	B1.1.4–6	Relate the structures of glucose, starch, glycogen and cellulose to their functions
3	B1.1.7–11	Explain glycoproteins in recognition, and how triglycerides form, vary and store energy
4	B1.1.12–13	Explain how phospholipids form bilayers, and why steroids can pass through membranes

The big picture: small units, big molecules, different jobs



Living things build huge molecules from a few kinds of small ones. **Carbohydrates** are built from sugars; many **lipids** are built from fatty acids and glycerol.

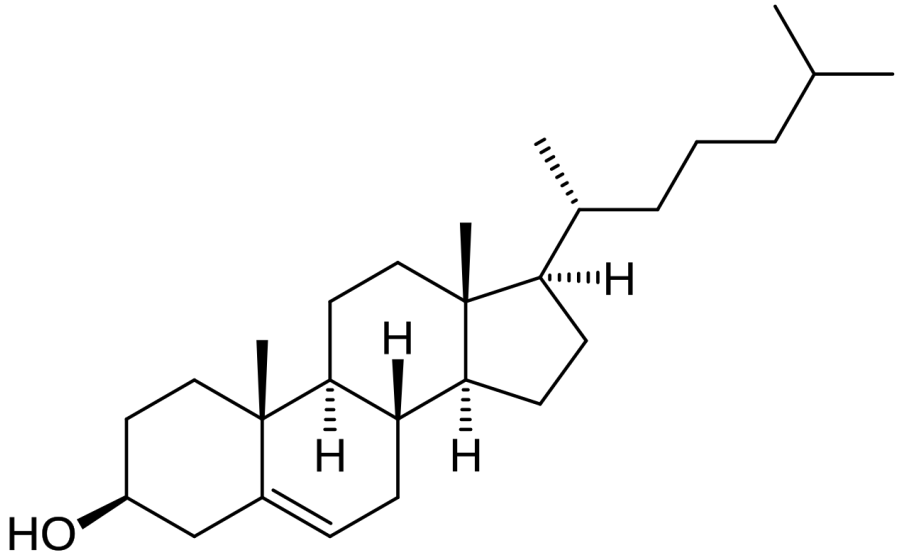
The same type of reaction, **condensation**, builds them all, and **hydrolysis** breaks them down again. What makes each molecule useful is its shape and its interaction with water.

Big question for this chapter

Starch, glycogen and cellulose are all made of glucose. Why is one a food store in plants, one a food store in animals, and one the strongest material in a plant?

Why carbon is the basis of life

B1.1.1



Cholesterol: rings and a branched chain. BorisTM, public domain.

A **covalent bond** forms when two atoms share a pair of electrons. Covalent bonds are strong and stable.

A carbon atom can form **four** covalent bonds: four single bonds, or a mix of single and double bonds. It can bond to other carbon atoms and to hydrogen, oxygen, nitrogen, phosphorus and sulfur.

This lets carbon form an almost endless variety of stable molecules of different sizes and shapes.

Key idea

No other element combines four bonds, strong carbon-to-carbon bonds and so many possible partners. That is why life is carbon-based.

Chains, branches and rings

B1.1.1

Arrangement	Example	Why the shape matters
Unbranched chain	Fatty acids; amylose in starch; cellulose	Straight chains can pack closely or lie side by side
Branched chain	Amylopectin in starch; glycogen	Many ends where units can be added or removed
Single ring	Glucose, ribose	Stable, compact units for building polymers
Several rings	Steroids such as cholesterol and testosterone	A flat, non-polar shape that fits into membranes

Try it

For each molecule in this chapter, say which arrangement of carbon atoms it has.

Condensation: joining monomers into polymers

B1.1.2



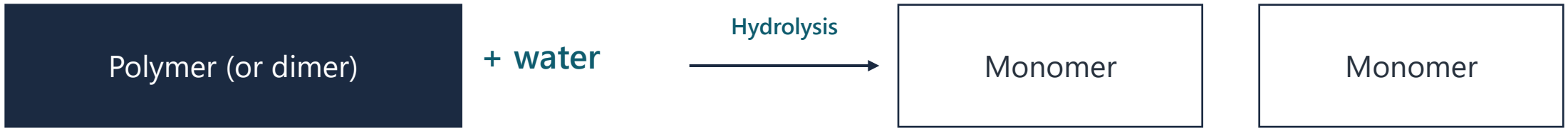
In **condensation**, an -OH from one monomer and an -H from another are removed as a molecule of **water**, and a new covalent bond joins the two monomers. Repeating this builds a **polymer**.

Building polymers needs energy, which comes from ATP, and enzymes catalyse each step.

Monomers	Polymer	Bond formed
Monosaccharides	Polysaccharide (starch, glycogen, cellulose)	Glycosidic bond
Amino acids	Polypeptide	Peptide bond
Nucleotides	Nucleic acid	Sugar-phosphate (phosphodiester) bond

Hydrolysis: breaking polymers into monomers

B1.1.3



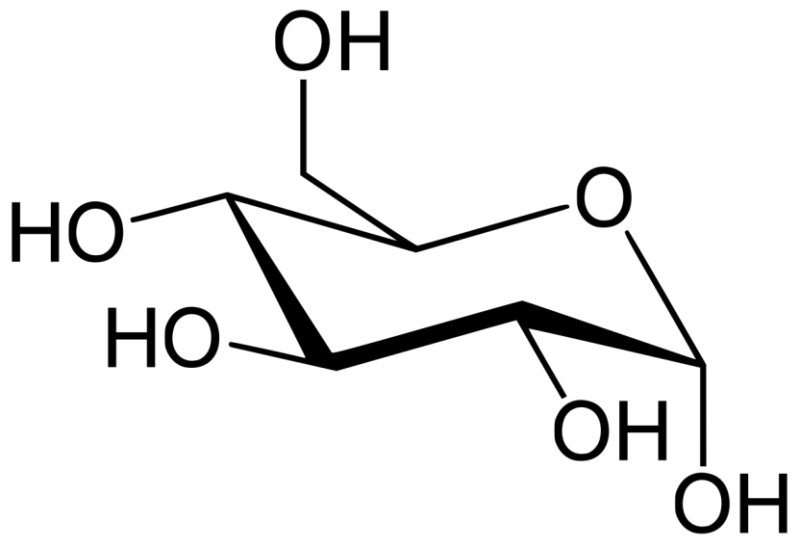
In **hydrolysis**, a water molecule is split and its -H and -OH are added back to the monomers on either side of a bond, breaking it. This is the reverse of condensation.

Digestion is hydrolysis, catalysed by enzymes, so that large food molecules become small enough to be absorbed.

Food polymer	Enzyme	Products
Starch	Amylase	Maltose, then glucose
Proteins	Proteases	Amino acids
Triglycerides	Lipase	Fatty acids and glycerol

Monosaccharides: single sugar units

B1.1.4



Alpha-glucose, a hexose. Yikrazuul, public domain.

Monosaccharides are the simplest carbohydrates. They dissolve in water and taste sweet.

In ring form, **pentoses** (such as ribose and deoxyribose) have 5 carbon atoms, and **hexoses** (such as glucose, fructose and galactose) have 6.

To identify one from a diagram, count the carbons. In a hexose ring, five carbons are in the ring with an oxygen, and the sixth sticks out above it.

Try it

Count the carbon atoms in the glucose diagram. Find the one carbon that is not part of the ring.

Why glucose is so useful

B1.1.4

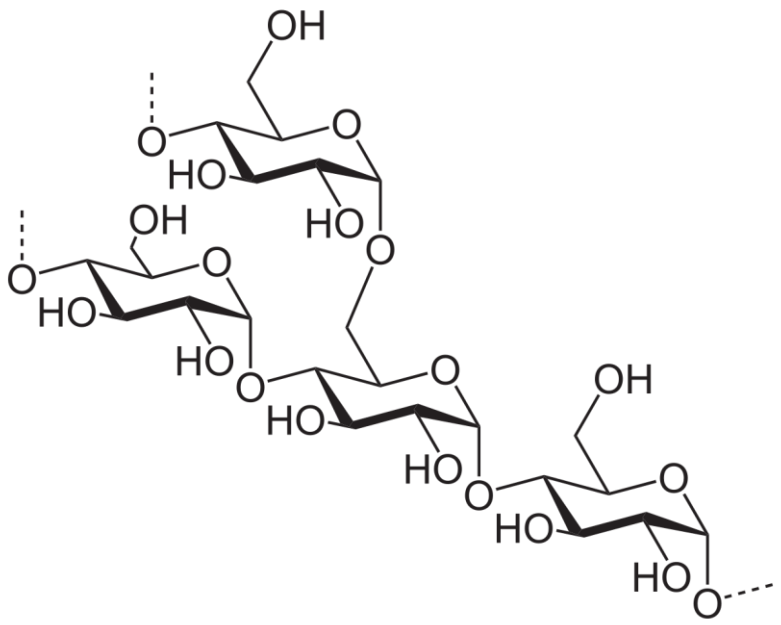
Property	Because	So glucose can
Soluble	Many -OH groups form hydrogen bonds with water	Dissolve in cytoplasm, blood plasma and phloem sap
Easily transported	It is small and dissolves	Be carried around the body to every cell
Chemically stable	Its ring structure does not react easily	Be stored and moved without breaking down
Yields energy when oxidized	Its bonds hold chemical energy	Be respired to release energy for ATP

Link

C1.2 Cell respiration: glucose is the main respiratory substrate.

Starch: the energy store of plants

B1.1.5



Part of amylopectin, in starch. NEUROtiker, public domain.

Starch is a polymer of **alpha-glucose** joined by glycosidic bonds. It has two forms:

Amylose is an unbranched chain that coils into a helix.

Amylopectin is branched: side chains join the main chain every 20 to 30 glucose units.

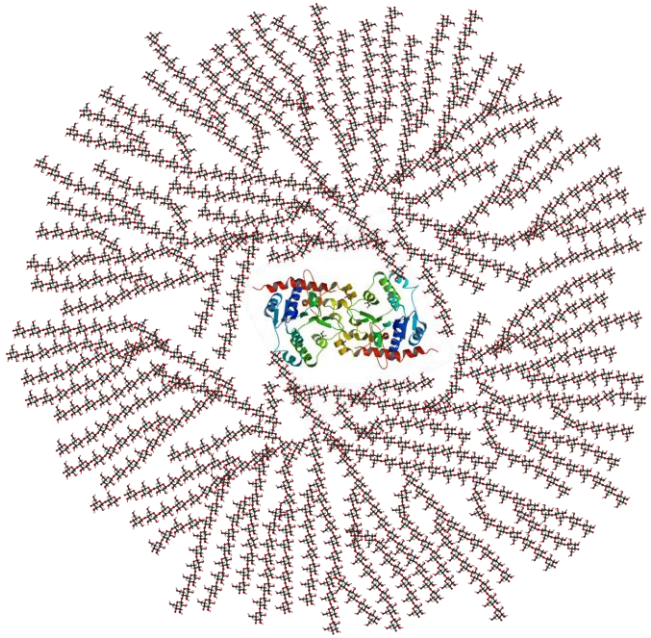
Plants store starch in grains in chloroplasts, and in storage organs such as potatoes and seeds.

Why the shape helps

Coiling and branching make starch compact. Being so large, it is insoluble, so it does not draw water into cells by osmosis.

Glycogen: the energy store of animals and fungi

B1.1.5



Branched glycogen. Mikael Haggstrom, public domain.

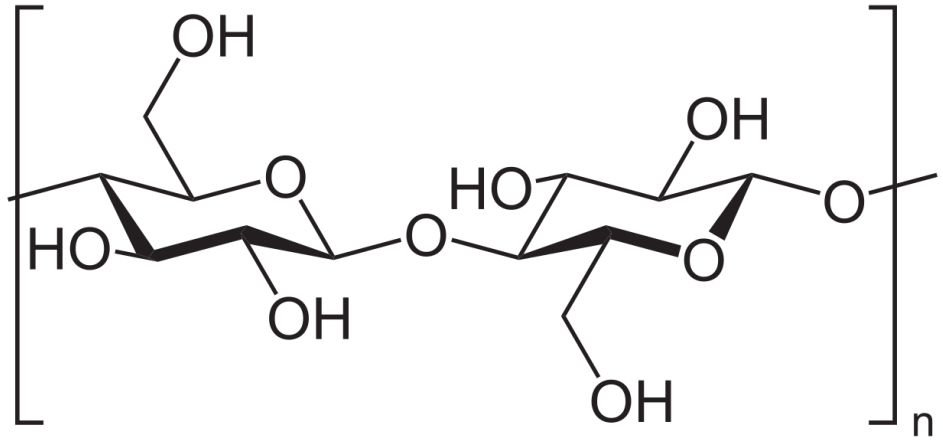
	Starch	Glycogen
Found in	Plants	Animals and fungi
Monomer	Alpha-glucose	Alpha-glucose
Branching	Amylose none; amylopectin some	Very highly branched
Stored in	Plastids in seeds, tubers and leaves	Liver and muscle cells

Key idea

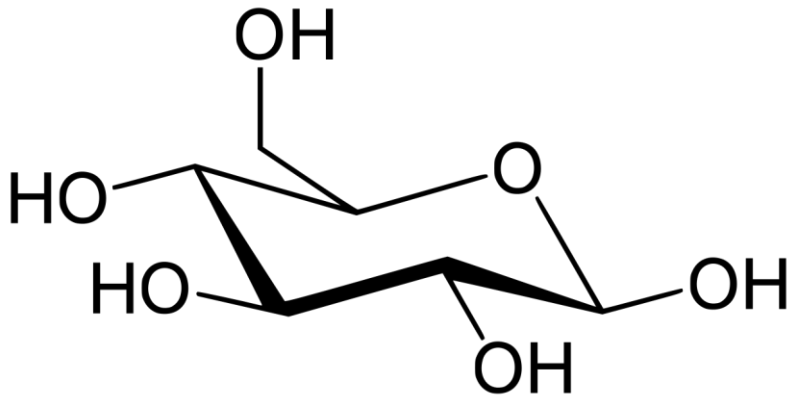
Every branch has an end where enzymes can add or remove glucose. Glycogen's many branches mean glucose can be released very quickly, which suits active animals.

Cellulose: a structural polysaccharide

B1.1.6



Part of a cellulose chain. NEUROtiker, public domain.



Beta-glucose. Yikrazuul, public domain.

Cellulose is made of **beta-glucose**. In beta-glucose the -OH on carbon 1 points up rather than down, so each glucose in the chain is turned 180 degrees from its neighbour. The result is a **straight**, unbranched chain.

Straight chains lie side by side, and **hydrogen bonds** link them into bundles called **microfibrils**, which are very strong in tension. They form plant **cell walls**.

Key idea

Cellulose and starch are both made of glucose. The small difference between alpha and beta glucose changes the shape of the polymer, and therefore its function.

Same monomer, different polymer

B1.1.5–6

	Starch (amylose)	Cellulose
Monomer	Alpha-glucose	Beta-glucose
Shape of chain	Coiled helix	Straight
Links between chains	None	Hydrogen bonds into microfibrils
Function	Energy store: compact, insoluble, easy to break down	Structure: strong cell walls
Digested by humans?	Yes, by amylase	No

Big question

Back to slide 3: alpha or beta glucose decides the shape of the chain, and the shape decides the job.

Check-in: carbon, polymers and carbohydrates

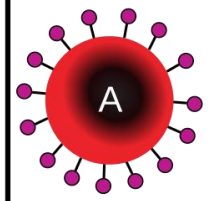
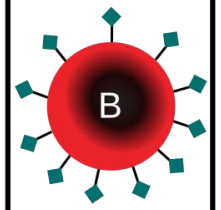
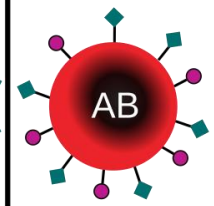
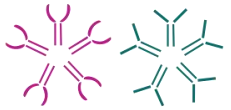
B1.1.1–6

1. State how many covalent bonds a carbon atom can form.
2. Distinguish between condensation and hydrolysis.
3. Identify a molecule with six carbon atoms in its ring form as a pentose or a hexose.
4. Explain two features of glycogen that suit it to energy storage.
5. Explain why cellulose forms straight chains but amylose does not.

How to use this

Answer without your notes, then compare with a partner.

Glycoproteins: name tags on cells

	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies in plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in red blood cell	A antigen	 B antigen	 A and B antigens	None

ABO antigens on red blood cells. InvictaHOG, public domain.

A **glycoprotein** is a protein with a short chain of sugars attached. Glycoproteins sit in the plasma membrane with the carbohydrate facing **outwards**.

Because the carbohydrate chains vary, other cells can **recognize** them. This is called **cell-cell recognition**: the immune system uses it to tell the body's own cells from foreign ones.

The **ABO blood groups** are an example: the antigens differ only in the sugar at the end of the chain.

ABO blood groups and transfusions

B1.1.7

Group	Antigens on red cells	Antibodies in plasma	Can receive blood from
A	A	Anti-B	A or O
B	B	Anti-A	B or O
AB	A and B	Neither	A, B, AB or O
O	Neither	Anti-A and anti-B	O only

Explain it

If a group A person receives group B blood, their anti-B antibodies bind the B antigens on the donated cells, which clump together (agglutinate). This is why blood must be matched before a transfusion.

Lipids: defined by not dissolving in water

B1.1.8

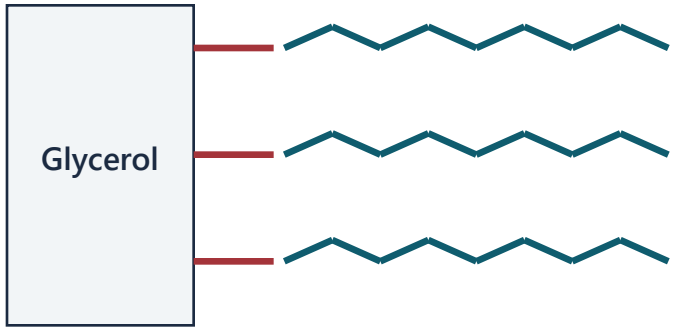
Lipids are a varied group of molecules defined by a property, not a structure: they dissolve in **non-polar solvents** such as ethanol but are only sparingly soluble in water. This is because they are made mostly of carbon and hydrogen, and C-H bonds are non-polar.

Lipid	Structure	Example of a use
Fats and oils	Triglycerides	Energy store; insulation (blubber)
Phospholipids	Glycerol, two fatty acids and a phosphate	Membranes
Waxes	Long fatty acids joined to long alcohols	Waterproof layer on leaves and feathers
Steroids	Four fused carbon rings	Hormones; cholesterol in membranes

Link

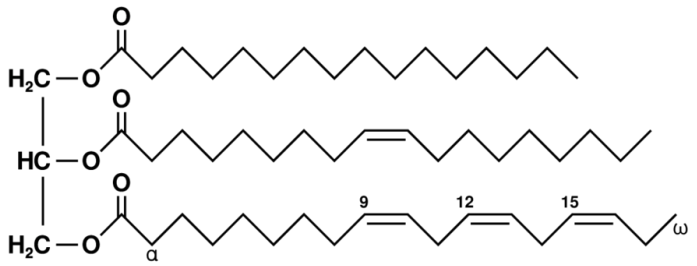
A1.1.5: some molecules work because they are hydrophobic.

Forming a triglyceride



Three fatty acid tails

Ester bonds (red)



A triglyceride. Wolfgang Schaefer, public domain.

A **triglyceride** forms when **three fatty acids** join to one **glycerol** by condensation.

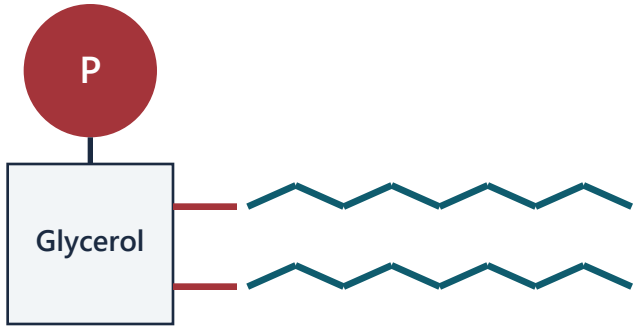
Each join forms an **ester bond** and releases a molecule of water, so three water molecules are released in all.

Activity

Draw a simplified triglyceride as on the left: a rectangle for glycerol and a zigzag line for each fatty acid. Then write the word equation: glycerol + 3 fatty acids → triglyceride + 3 water.

Forming a phospholipid

B1.1.9



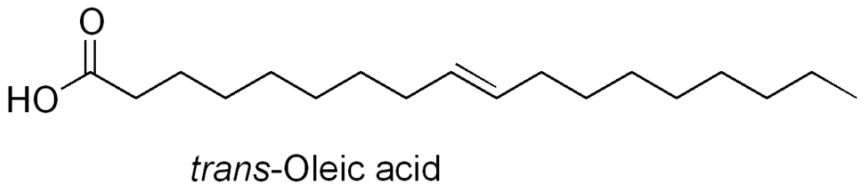
Phosphate head, glycerol, two tails

	Triglyceride	Phospholipid
Glycerol	One	One
Fatty acids	Three	Two
Phosphate group	None	One, in place of a fatty acid
Main role	Energy storage	Membranes

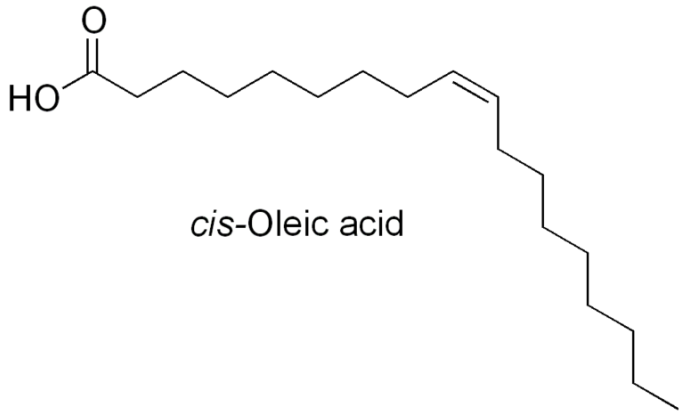
In a **phospholipid**, glycerol is joined to **two** fatty acids and a **phosphate group**. The phosphate end is charged and **hydrophilic**; the two fatty acid tails are **hydrophobic**. This difference is the key to how membranes form (slide 25).

Saturated and unsaturated fatty acids

B1.1.10



trans-Oleic acid



cis-Oleic acid

Shapes of unsaturated fatty acids. Edgar181, public domain.

A **saturated** fatty acid has only single bonds between carbons, so it holds the maximum number of hydrogen atoms. Its chain is straight.

An **unsaturated** fatty acid has one (**mono**unsaturated) or more (**poly**unsaturated) C=C double bonds. In natural **cis** fats, each double bond puts a **kink** in the chain.

Type	Effect on packing and melting point
Saturated	Straight chains pack closely: higher melting point, solid fats
Unsaturated (cis)	Kinks stop close packing: lower melting point, liquid oils

Why it matters

Plants store oils, which stay liquid in the cold. Mammals and birds store fats, which are liquid at body temperature but solid at room temperature.

Triglycerides in adipose tissue

B1.1.11



Walruses, with thick blubber. Andrew Trites, NOAA, CC BY 2.0.



Adipose tissue. Chalova.lina, CC BY 4.0.

Adipose tissue is made of cells filled with triglyceride droplets.

It is a **long-term energy store**: triglycerides are insoluble and hold about twice as much energy per gram as carbohydrate.

It is a **thermal insulator**, because fat conducts heat poorly.

Link to habitat

Blubber is thickest in endotherms in cold water, which conducts heat away quickly (A1.1.6). Animals in hot climates often store fat in one place, such as a camel's hump, so it does not insulate the whole body.

Lipids or carbohydrates for storing energy?

B1.1.11

	Triglycerides	Glycogen and starch
Energy per gram	About 37 kJ: twice as much	About 17 kJ
Mass for the same energy	Low: good for animals that move or fly	About twice as heavy (more if stored with water)
Speed of release	Slow to mobilize	Fast to mobilize
Respiration	Aerobic only	Aerobic or anaerobic
Used for	Long-term storage	Short-term storage

Guiding question

Lipids store more energy per gram; carbohydrates release it faster and can be used without oxygen.

Working with data: fatty acids and melting point

B1.1.10

All four fatty acids have 18 carbon atoms. Melting points are published values, rounded.

Fatty acid	C=C double bonds	Melting point / °C
Stearic acid	0	70
Oleic acid	1	13
Linoleic acid	2	-5
Linolenic acid	3	-11

1. Describe the relationship between the number of double bonds and the melting point.
2. Explain this relationship.
3. Which fatty acids would be liquid in a fish living at 4 °C?
4. Suggest why seeds of plants from cold climates contain more unsaturated oils.

Check-in: glycoproteins and triglycerides

B1.1.7–11

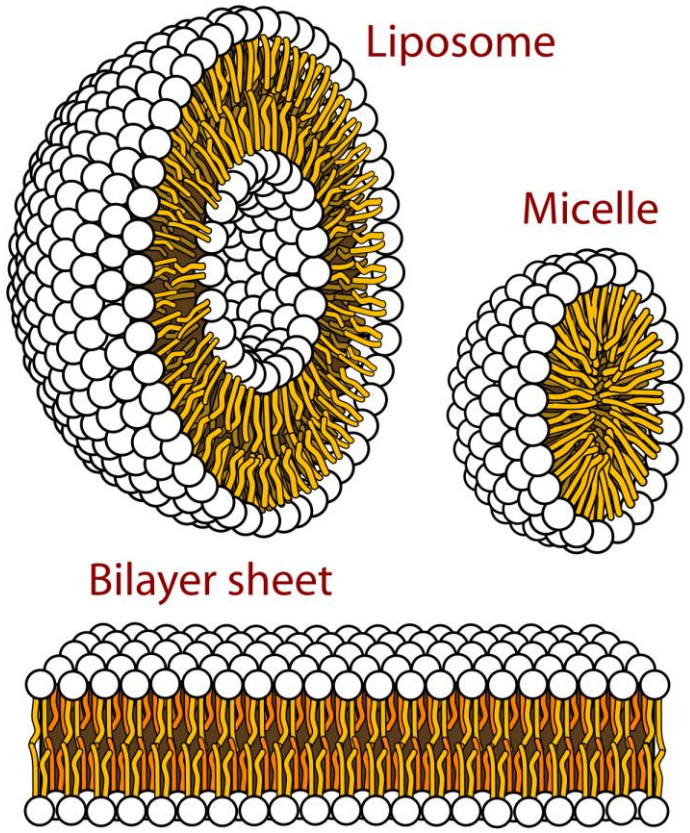
1. Outline how glycoproteins allow cells to recognize each other.
2. State the number of water molecules released when a triglyceride forms.
3. Distinguish between a saturated and an unsaturated fatty acid.
4. Explain why triglycerides are better than glycogen for long-term energy storage.
5. Explain why blubber is thickest in animals living in cold water.

How to use this

Write full sentences, then mark with a partner.

Phospholipids form bilayers by themselves

B1.1.12



Phospholipids in water. LadyofHats, public domain.

A phospholipid has a **hydrophilic** phosphate head and two **hydrophobic** tails. A molecule with both kinds of region is **amphipathic**.

In water, the heads are attracted to water and the tails are not. The molecules arrange themselves in a double layer, a **bilayer**, with the heads facing the water on both sides and the tails hidden in the middle.

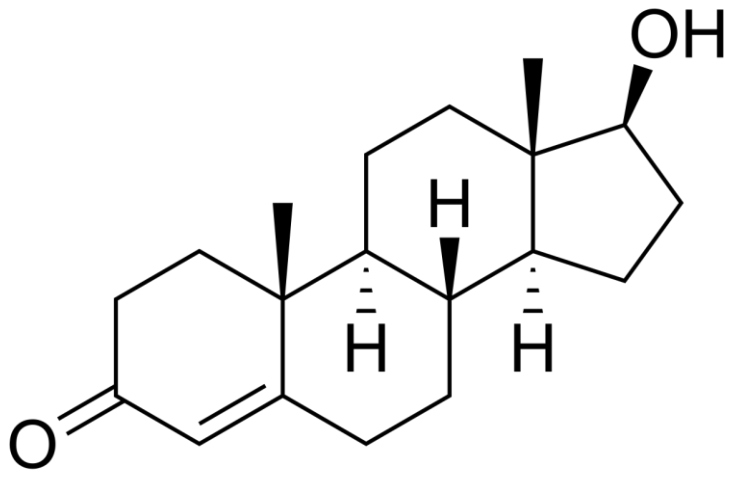
This happens without any energy input, and it forms the basis of every **cell membrane**.

Link

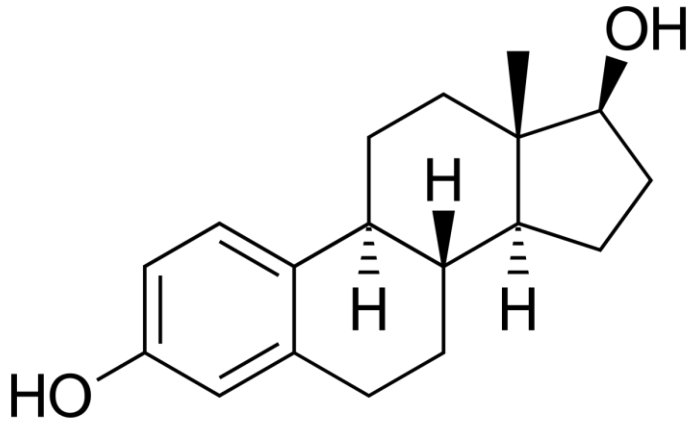
B2.1 Membranes; A2.1 HL: self-assembling bilayers and the origin of the first cells.

Steroids can pass through membranes

B1.1.13



Testosterone. NEUROtiker, public domain.



Oestradiol. NEUROtiker, public domain.

Steroids are lipids built from **four fused rings** of carbon atoms: three rings of six carbons and one of five.

They are mostly **non-polar**, so they can dissolve into the hydrophobic core of a membrane and pass straight through it.

So steroid hormones such as **oestradiol** and **testosterone** can enter their target cells and bind to receptors **inside** the cell.

Try it

Identify steroids from molecular diagrams by looking for the pattern of four rings. You do not need to learn the side groups.

Common mistakes to avoid

Not this	But this
Condensation uses water.	Condensation releases water; hydrolysis uses it.
Cellulose is made of alpha-glucose.	Cellulose is made of beta-glucose; starch and glycogen use alpha-glucose.
A triglyceride is three glycerols and a fatty acid.	It is one glycerol and three fatty acids.
Unsaturated fats have more hydrogen.	Saturated fatty acids hold the most hydrogen; unsaturated ones have C=C double bonds.
Phospholipid heads are hydrophobic.	Heads are hydrophilic; the fatty acid tails are hydrophobic.
Steroids enter cells through protein channels.	Steroids are non-polar, so they pass through the lipid bilayer itself.

Check that you can...

1. **Outline** why carbon can form so many different compounds
2. **Distinguish** between condensation and hydrolysis, with examples
3. **Identify** pentoses and hexoses, and **explain** why glucose suits its roles
4. **Compare** starch, glycogen and cellulose, relating structure to function
5. **Outline** the role of glycoproteins, using ABO blood groups
6. **Draw** simplified triglycerides and phospholipids, and **explain** how they form
7. **Explain** the effect of saturation on melting point, and why triglycerides store energy
8. **Explain** how phospholipids form bilayers and why steroids cross membranes

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the type of reaction that links two monosaccharides, and the molecule that it releases.
2. Give two ways in which the structure of glycogen suits it to energy storage.
3. Explain why a phospholipid is described as amphipathic.

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